

Lorenzo Ghessi Master's student in Automation and Control Engineering

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ABOUT ME

MSc Automation & Control Engineering student (PoliMi + Chalmers Erasmus+), graduating April 2027. Currently developing a model-mediated teleoperation system with communication delay compensation at Leonardo S.p.A. Background spans ROS/C++ software development, PLC commissioning on industrial machines, and hands-on cobot programming.

EXPERIENCE

Leonardo S.p.A, Robotics Research Intern | Model-Mediated Teleoperation 07/2026 – Present | Genoa, Italy

- Designed and implemented a model-mediated teleoperation architecture in ROS/C++ to compensate for variable communication delays, enabling stable robotic manipulation under realistic network latency conditions.
- Developed and validated control algorithms for object manipulation with delayed feedback, integrating hardware-in-the-loop testing on a real robotic system.

Assolombarda, Laboratory Coordinator | Robotics Educational Lab 09/2021 – Present | Monza

- Managed a team of 10 facilitators across shift planning and technical operations, maintaining full uptime of ABB YuMi cobots and Lab 4.0 equipment.
- Designed new didactic modules integrating ABB RobotStudio and cobot programming, expanding the curriculum adopted by Assolombarda for STEM outreach programmes.
- Delivered hands-on robotics workshops to 1000+ middle and high school students across 4+ years, achieving consistently high participant satisfaction.

Delta Service Automation, Engineering Intern | PLC & HMI 09/2024 – 10/2024 | Vaprio d'Adda

- Debugged and optimized PLC/HMI software (DELTA) on automatic labeling and product-tracking machines, directly reducing machine downtime during production runs.
- Revised electrical schematics, authored technical maintenance manuals, and modeled 3D mechanical assemblies for production line integration in SolidWorks.

PROJECTS

Flexible Joint Control (MATLAB/Simulink) 02/2026 – 06/2026

Experimental parameter identification (friction, spring stiffness) and synthesis of frequency-domain and state-space controllers (LQR, H ∞ , MRAC, SMC) for precise tip positioning under uncertainty.

Trajectory control for a Mobile Robot (ROS/C++), Milan, Italy 10/2025 – 11/2025

Built a Turtlebot simulator in Linux/ROS Noetic (Boost Odeint); designed a PI tracking controller with feedback linearization and velocity feed-forward.

Drone Real-Time Control (C/C++, FreeRTOS), Gothenburg, Sweden 04/2025 – 06/2025

Developed dynamic model in Simscape, synthesized LQR controller in MATLAB/Simulink, and implemented control code on real hardware using an RTOS architecture.

Industrial Robotic Cell 4.0 (PLC/HMI, Industrial Robotics), 03/2021 – 06/2021

Sesto San Giovanni, Italy
Designed a pick-and-place cell integrating ABB RobotStudio, Siemens S7-1200 PLC, HMI, and a computer vision system for quality inspection.

EDUCATION

Polytechnic University of Milan, MSc Automation and control engineering 09/2024 – Present | Milan, Italy

Chalmers University of Technology, Erasmus 01/2025 – 06/2025 | Gothenburg, Sweden

Polytechnic University of Milan, BSc in Automation and control engineering 09/2021 – 07/2024 | Milan, Italy

SKILLS AND CERTIFICATIONS

Software

ROS/ROS2, MATLAB/Simulink, ABB RobotStudio, SolidWorks, Linux (Ubuntu), AutoCAD

Programming

C/C++, Python, PLC (Ladder, ST, SFC)

Core competencies

Multivariable control, Robot programming (ABB RAPID), Industrial commissioning, Computer Vision, Real-time systems (FreeRTOS), Teleoperation

Languages

Italian (native), English (C1 – IELTS 7.0), Spanish (elementary), German (beginner, A1)